Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/25/2025
PI Name:	Fornace, Albert M.D.		
Project Title:	Radiation Carcinogenesis by GCRsim in Animal Models for High Priority Cancer Types (NSCOR)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	http://www9.georgetown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/25/2022	**End Date:**	05/24/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/24/2025 per NSSC information (Ed., 5/2/24). NOTE: End date changed to 05/24/2024 per S. Mack/JSC. Original end date was 05/24/2023 (Ed., 5/22/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shubhankar, Suman Ph.D. (Georgetown University) Shay, Jerry (University of Texas Southwestern Medical Center at Dallas) Brenner, David (Columbia University)		
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Task Description:	Considering the high spontaneous incidence of gastrointestinal (GI) and lung cancer, an even modest increase by space radiation exposure could have a significant effect on astronauts' health risk estimates during and after long-duration manned space flights. However, there is substantial uncertainty for GI and lung cancer risk estimation from space radiation due to the lack of in vivo human data. The overarching goal of the current proposal is to specifically investigate cellular and molecular hypothesis-driven mechanisms in relevant mouse models that will contribute to carcinogenic risk estimates of GI and lung cancer after exposure to space radiation beams prioritized by NASA's Human Research Program (HRP). Specifically, this proposal builds on the data generated from previous NSCOR studies and seeks to determine incidence, tumor frequency, and grade as well as identify molecular perturbations in the target cells associated with GI and lung tumorigenesis through monitoring of changes in the tumor number, histology, gene/protein expression, and metabolome of the proposed model system after exposure to space radiation type beams. Having found modest effects in earlier studies with protons and considering the importance of assessing the effects of mixed beams, we have chosen high priority Galactic Cosmic Radiation (GCR) type beams for comparison with low-LET (gamma-rays) radiation in our proposed studies. The full-spectrum (33-ion) acute (single dose) and protracted (up to 6 weeks) GCRsim beams closely mimic the mixed particle radiation environment in deep space. Our overall hypothesis is that GCRsim exposure is more carcinogenic due to important qualitative differences from low-LET gamma rays. The overall objective of this proposal is to acquire quantitative and qualitative cancer data after full-spectrum GCRsim exposure to model the relative risk of GI and lung tumorigenesis and compare it to gamma radiation where human epidemiologic data are available. Our proposed Aims are: Aim 1. Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2. Understanding molecular signaling associated with space radiation-induced GI-tumorigenesis and functional alterations. Aim 3. Quantitatively and qualitatively compare effects of GCR-type irradiations on lung cancer initiation and progression in normal as well as lung cancer susceptible mice. Aim 4. Development of mathematical modeling for GI and lung cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	According to estimations from the American Cancer Society (ACS), the lifetime risk of developing colorectal (1 in 23 for men and 1 in 25 for women) and lung (1 in 16 for men and 1 in 17 for women) cancer is significantly high in the US population. Considering the high prevalence of GI and lung cancer, an even slight increase in the incidence of GI and lung cancer in astronauts due to space radiation exposure is a critical health concern and has a significant impact on the planning of upcoming manned deep space exploration missions. In addition to cancer risk estimation studies after space radiation exposure, investigations on the persistence of oxidative and inflammatory stress after space radiation and its role in GI and lung cancer development may provide insight into mutagenic processes affecting genome integrity and carcinogenesis. The significance and deliverable of this project are to improve the estimates of GI and lung cancer risk after simulated galactic cosmic radiation (GCR) exposure and to identify plausible targets for the development of medical countermeasures (MCM).
Task Progress:	Pointwise progress made for Aim/Projects 1–4 which is summarized below: 1. We investigated the dose-response effects (10–75 cGy range) and dose-rate effects (chronic vs. acute) of the full-spectrum 33-ion GCRsim on GI tumorigenesis in male and female Apc1638N/+ mice. Both acute and chronic GCRsim exposures at 25, 50, and 75 cGy significantly increased GI tumorigenesis in both sexes compared to controls. Furthermore, we observed a significant dose-dependent increase in GI tumorigenesis across all GCRsim doses, with male mice demonstrating higher susceptibility than females. 2. Acute GCRsim exposure at 75 cGy resulted in a >2-fold increase in intestinal tumors, compared to sham controls. Chronic 75 cGy exposure (delivered in 6 weeks at 2.08 cGy/day, 6 days in a week) led to the highest tumorigenic response, which is comparable to tumorigenesis after same dose of acute GCRsim (delivered in ~30 min), indicating no dose rate sparing. 3. We also investigated the tumorigenic potential of Low-LET (protons + helium) vs. High-LET (HZE only) components of the GCRsim on GI-tumorigenesis in male Apc1638N/+ mice. High-LET (HZE) particles showed significantly higher tumorigenic potential compared to low-LET (protons + helium). Contribution to total tumorigenesis per unit dose in a mixed field GCRsim originating from the HZE only component was ~20 times higher than from protons + helium, underscoring the dominant role of high-LET radiation in driving intestinal tumor formation. 4. GCRsim exposure caused more pronounced activation of the cGAS-STING pathway compared to γ-rays, highlighting the unique biological effects of densely ionizing radiation. Acute GCRsim exposure significantly activated cGAS-STING signaling in the mouse intestine, surpassing effects seen with γ-ray exposure. Immunoblot analysis revealed elevated STING, pTBK1, and pIKKα levels, indicating robust downstream activation. Immunohistochemistry demonstrated increased nuclear pIRF3, pIRF7, and p65, implicating interferon signaling and NF-κB-mediated inflammation. In conclusion, GCRsim exposure may lead to sustained inflammation, tissue damage, gastrointestinal dysfunction, and increased cancer risk, and highlights the role of cGAS-STING signaling in driving intestinal damage due to space radiation. 5. Disruption of nutrient absorption and barrier integrity increases susceptibility to inflammation, systemic effects, and gastrointestinal dysfunction during prolonged space missions. Upregulation of glucose transporter genes (Scl2a2, Scl2a5, Scl5a1) and gut hormone pathways (Cck, Gip), suggests compensatory mechanisms for energy homeostasis. Downregulation of lipid transporters (Npc1, Npc1l1) indicated selective impairment in lipid absorption. Reduced serum citrulline levels in GCRsim-exposed mice indicated compromised intestinal epithelial health. Elevated serum I-FABP levels highlighted significant intestinal injury and barrier dysfunction. These findings underscore the importance of targeted interventions to mitigate space radiation-induced intestinal dysfunction and systemic effects in astronauts on long-duration missions. 6. Prolonged crewed spaceflight poses health risks due to exposure to galactic cosmic radiation, with previous research at the NSRL indicating that HZE ion components of GCR lead to persistent inflammation, increased mutations, and elevated cancer rates. Utilizing the 33-beam GCRsim at NSRL, experiments on a lung cancer susceptible mouse model i.e., K-rasLA-1 mice showed increased lung adenocarcinoma incidence with higher doses and prolonged exposures, particularly when neutron exposure was added. We interpret these findings to suggest that the risks of carcinogenesis are heightened with doses anticipated during a round trip to Mars, and this risk is magnified when coupled with extra neutron exposure on the Martian surface. We also observed that risks were reduced when the 33-beam GCRsim exposures were conducted at a high dose rate compared to a low dose rate.

	7. We demonstrate a radioprotective effect of metformin through an indirect modulation of gene expression involved in cellular detoxification rather than its effects on mitochondria. Administration of metformin (0.5 mM) showed radioprotective effects in vitro on BJ human fibroblasts, increasing DNA damage repair and increasing SOD1 expression in the nucleus. Importantly, metformin (200 mg/kg) pre-administration for only 3 days in wild type 129/sv mice, decreased the formation of micronuclei in bone marrow cells and DNA damage in colon and lung tissues compared to control irradiated mice at sub-lethal and lethal doses, increasing the overall survival fraction by 37% after 10 Gy total body irradiation. We next pre-treated with metformin and then exposed 129/sv mice, to the 33-beam GCRsim. We found metformin pre-treatment decreases the presence of bone marrow micronuclei and DNA damage in colon and lung tissues and an increase of 8-oxoguanine DNA glycosylase-1 (OGG1) expression. 8. A mechanistic model incorporating targeted effects (TE) and non-targeted effects (NTE) was developed for intestinal tumorigenesis in Apc1638N/+ mice. The model revealed that simple effect additivity (SEA) does not accurately predict cancer risks for mixed space radiation types with non-linear dose-response relationships. 9. The Incremental Effect Additivity (IEA) methodology was applied to predict the tumorigenic effectiveness of mixed radiation exposures (protons, heavy ions, and gamma rays). A new dataset from GCRsim exposures (acute and chronic) is being used to refine the TE+NTE model, offering a detailed and realistic assessment of space radiation effects and individual component contributions.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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